

Synthetic data

How fake data can be useful

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About me

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Nowadays, more data is collected than ever
before^I



The entire world is digital

- News articles
- Libraries
- Soil data
- Economic indices
- Social media
- Your online web-browsing behavior
- CBS data
- Research data



Data is a gold mine

Embed from Getty Images



Researchers need rich data!

- Tracking pandemics in real time
- Statistics Netherlands (CBS) population registry
- FIRMBACKBONE: Dutch longitudinal data of registered companies



Companies also benefit from rich data

- Stock market predictions
- Predicting trends
- i. According to **Statista**, approximately 163 zettabytes (163.000.000.000.000 GB) of data was created in the year 2025
- Almost all webshops evaluate their marketing success alone.



All that glisters is not gold

(Open) data puts us at risk!

5 minute assignment: Think about risks of open data for (1) individuals, and (2) companies.

(Open) data risks

For individuals: identity theft, fraud, stalking, discrimination, blackmail, deepfakes

For companies: leakage of trade secrets, customer targeting by competitors, copyright or IP issues, reputational damage

For the general public: misinformation, surveillance, unequal power, loss of trust, drowning in low-quality data, and environmental costs of storing and processing massive datasets

Some known data breaches

- Netflix prize data
- Strava runs of military personnel
- And of course all kinds of hacks

Connecting different data sources may amplify risks!



Should we stop collecting and sharing data?

We need better protection strategies!



Data protection strategies

- No data sharing (wasteful)
- (Physical) research data centers
- Remote access servers
- Disseminating aggregate statistics
- Data perturbation
- Synthetic data
- Open data dissemination (harmful)



Data anonymization

What is identifying information?



Synthetic data

Synthetic data are data that are generated from a model or algorithm, with the aim of mimicking (some aspects of) the

- I. ~~Fake data, generated data~~ Fake data, generated data, digital twins. As opposed to real world, collected data.





The synthetic data machine

Don't you ever!

Learning the structure of a data set

What features are important?



Distinguishing signal from noise



Example: The U.S. Current Population Survey

- 5000 individuals
- Variables: tax, income, age, educ, marital status, race, sex, social security



Univariate distributions

Multivariate distributions

Privacy risks



Identity disclosure

The ability to identify an individual from the synthetic data



Attribute disclosure

The ability to learn an attribute (e.g., some characteristic) of an individual from the synthetic data

The privacy-utility trade-off

Utility versus privacy

Utility

- Can I do the same things with the synthetic data as with the real data?
- Are the synthetic data characteristics the same as the observed data characteristics?

Privacy

- Can I reproduce the original data from the synthetic data?
- Can I determine whether a person was part of the original data?
- Can I estimate person-specific characteristics with great precision?



How much does the synthetic data look like the real data?

Perfect imitation



In principle, you always lose information when creating data. The question is 'how much information do you sacrifice in order to protect privacy?'



Tools for generating synthetic data

Many software programs exist (mainly in programming languages like **R** and **python**)

- For tabular data: specify model for each variable
- For images: deep learning model for the entire image
- For text, videos: deep learning models for sequences



Using synthetic data

We want to do the same things with synthetic data that we could do with the real data, without the privacy risks related to the real data.



Synthetic data use-cases

Dummy data

Data for education

Data for testing and model-building

Replication packages

Novel research?



Advantages of synthetic data

Privacy can be well protected

- No real records anymore
- Information that goes into the synthesis model can be bounded
A large amount of information can be preserved
- Model can be enriched with additional information

Possible extensions

Encode prior or external information

- Different data sources

Borrow strength from different domains



Disadvantages of synthetic data

Quality of synthetic data depends entirely on the synthesis model; and can be hard to evaluate

- Too simple models might omit important data characteristics
- Complex sythesis models obscure what information is reproduced in the synthetic data

Synthetic data can live a life of it's own

Assignment

1. Discuss a case where you may be harmed by your own published (real or hypothetical) data
2. Discuss whether synthetic data can serve as a protection method, or whether something else is required
3. Evaluate potential downsides of the protection methods

Approximately 500 words

Questions?

Feel free to reach out: t.b.volker@uu.nl
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Speaker notes

